EVERY OBJECT OF THE DERIVED CATEGORY OF AN INVERSE SYSTEM OF RINGS IS REPRESENTED BY A SYSTEM OF K-FLAT COMPLEXES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes '(details omitted)' inside the counted proof span and has the typo 'K_{n +1}^\bullet' with a missing space before the plus in the last paragraph; both are reproduced unchanged. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Representation of an object of the derived category of an inverse system by a system with surjective transition maps. invoked at proof lines 23824-23826. Is the reduction that makes the fibred-product construction available.
- Existence of K-flat resolutions of complexes. invoked at proof lines 23827-23829 and 23853-23854. Supplies the K-flat complexes at each stage.
- Termwise surjectivity and the quasi-isomorphism from a fibred product of complexes to the target. used at proof lines 23846-23852. Is the induction step that produces the next compatible stage.

Source reference. Original lemma 14380: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14380–14380. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 23780: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23780–23780. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let (A_n) be an inverse system of rings. Every $K \in D(Mod(\mathbf{N}, (A_n)))$ can be represented by a system of complexes $(K_n^\bullet)$ such that each $K_n^\bullet$ is K-flat.*

Proof. First use Lemma 23780 to represent K by a system of complexes $(M_n^\bullet)$ such that all the transition maps $M_{n+1}^\bullet \to M_n^\bullet$ are surjective. Next, let $K_1^\bullet \to M_1^\bullet$ be a quasi-isomorphism with $K_1^\bullet$ a K-flat complex of A_1-modules (Lemma 14380).

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Suppose we have constructed $K_n^\bullet \to K_{n-1}^\bullet \to \ldots \to K_1^\bullet$ and maps of complexes $\psi_e : K_e^\bullet \to M_e^\bullet$ such that

$$\begin{array}{ccc} K_{e+1}^\bullet & \longrightarrow & K_e^\bullet \\ \downarrow & & \downarrow \\ M_{e+1}^\bullet & \longrightarrow & M_e^\bullet \end{array}$$

commutes for all $e < n$. Then we consider the diagram

$$\begin{array}{ccc} C^\bullet & \dashrightarrow & K_n^\bullet \\ \vdots & & \downarrow{\scriptstyle \psi_n} \\ M_{n+1}^\bullet & \xrightarrow{\varphi_n} & M_n^\bullet \end{array}$$

in $D(A_{n+1})$. As $M_{n+1}^\bullet \to M_n^\bullet$ is termwise surjective, the complex $C^\bullet$ fitting into the left upper corner with terms

$$C^p = M_{n+1}^p \times_{M_n^p} K_n^p$$

is quasi-isomorphic to $M_{n+1}^\bullet$ (details omitted). Choose a quasi-isomorphism $K_{n+1}^\bullet \to C^\bullet$ with $K_{n+1}^\bullet$ K-flat. Thus the lemma holds by induction. $\square$